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ExoTiC-LD: thirty seconds to stellar limb-darkening coefficients

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Summary

Stellar limb darkening is the observed variation in brightness of a star between its centre and edge (or limb) when viewed in the plane of the sky. Stellar brightness is maximal at the centre and then decreases radially and monotonically towards the limb – hence the term "limb darkening". This effect is crucial for finding and characterising planets beyond our Solar System, known as exoplanets, as these planets are often studied when crossing in front of their host stars. As such, limb darkening is directly linked to the exoplanet signals. Limb darkening is typically modelled by one of various functional forms, as outlined in Claret (2000) and Sing (2010), and the coefficients of these functions is what `ExoTiC-LD` is designed to compute. A wide variety of functional forms are supported, including those benchmarked by Espinoza & Jordán (2016) as well as reparameterisations suggested by Kipping (2013).

Statement of need

Stellar limb darkening depends on the type of star, the wavelengths of light being observed, and the sensitivity of the instrument/telescope performing the observation. Therefore, to compute limb-darkening coefficients requires a frustrating amount of "data admin". In brief, one starts with a search through grids of stellar models to find a good match with the science target in metallicity, effective temperature, and surface gravity. Then, one must retrieve the wavelength-dependent sensitivity of the employed instrument, process all these data into the correct form, and then finally compute the limb-darkening coefficients.

In `ExoTiC-LD` we have done all of the heavy lifting for the user, making the process as fast and frictionless as possible. A user simply has to `pip install` the code and the relevant data will be automatically downloaded at runtime and the limb-darkening coefficients computed. In particular, all the stellar and instrument data has been pre-processed and homogenised. Additionally, the stellar model grids have been stored as tree structures, enabling an efficient search for good matches and helpful warnings to the user.

`ExoTiC-LD` thus far has predominantly been utilised in the study of exoplanet atmospheres, helping to facilitate the study of Jupiter-like (e.g., Alderson et al., 2023; Grant et al., 2023), Neptune-like (Radica et al., 2024; e.g., Roy et al., 2023), and Earth-like exoplanets (Kirk et al., 2024; e.g., Moran et al., 2023). It has also been incorporated into the popular open-source JWST data reduction and analysis pipeline, called `Eureka!` (Bell et al., 2022).

Currently, the stellar models supported are PHOENIX (Husser et al., 2013), kurucz (Kurucz, 1993), stagger (Magic et al., 2015), and MPS-ATLAS (NM Kostogryz et al., 2022; N. Kostogryz et al., 2023). There are also options to provide custom data if the user has their own stellar models or instrument data. Other available codes include those referenced in Southworth (2015), Parviainen & Aigrain (2015), Morello et al. (2020), and Bourque et al. (2021).

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